DISASTER MANAGEMENT FRAMEWORK AND SENDAI

DISASTER-RISK EQUATION → HAZARD-TO-DISASTER THRESHOLD → RISK-REDUCTION LADDER → BUILD BACK BETTER LOOP → DOMESTIC ARCHITECTURE FIREWALL → TRAINING-RESPONSE SPLIT

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 DISASTER-RISK EQUATION

DISASTER-RISK EQUATION | POTENTIALLY DAMAGING PROCESS OR EVENT | PEOPLE ASSETS SYSTEMS IN HARM'S WAY | SUSCEPTIBILITY TO DAMAGE | RESOURCES AND ABILITY TO MANAGE RISK | INTERACTION ACROSS A STATED PLACE AND TIME

SYSTEM / LEVEL / DEFINITION

- HAZARD → potentially damaging process or event
- EXPOSURE → people assets systems in harm's way

STRUCTURE-FUNCTION MECHANISM

- VULNERABILITY → susceptibility to damage
- CAPACITY → resources and ability to manage risk

SCALE / EVIDENCE LIMIT

- RISK → interaction across a stated place and time
- HAZARD → potentially damaging process or event

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RISK → interaction across a stated place and time.

01

STAGE 01 HAZARD-TO-DISASTER THRESHOLD

HAZARD-TO-DISASTER THRESHOLD | HAZARD OCCURS | PHYSICAL OR HUMAN-INDUCED EVENT | EXPOSED SYSTEM | CONTACT WITH PEOPLE ASSETS OR ECOSYSTEMS | VULNERABILITY AND LOW CAPACITY | LOSSES AMPLIFY | COPING ABILITY SEVERELY TESTED

HAZARD OCCURS → physical or human-induced event

EXPOSED SYSTEM → contact with people assets or ecosystems → VULNERABILITY AND LOW CAPACITY → losses amplify → COPING ABILITY SEVERELY TESTED → serious disruption

SYSTEM / LEVEL / DEFINITION

- HAZARD OCCURS → physical or human-induced event
- EXPOSED SYSTEM → contact with people assets or ecosystems

STRUCTURE-FUNCTION MECHANISM

- VULNERABILITY AND LOW CAPACITY → losses amplify
- COPING ABILITY SEVERELY TESTED → serious disruption

SCALE / EVIDENCE LIMIT

- DISASTER → not every hazard crosses this threshold
- HAZARD OCCURS → physical or human-induced event

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DISASTER → not every hazard crosses this threshold.

02

STAGE 02 RISK-REDUCTION LADDER

RISK-REDUCTION LADDER | AVOID CREATION OR EXISTENCE OF RISK WHERE POSSIBLE | REDUCE ADVERSE IMPACT OF REMAINING RISK | READY PLANS WARNINGS SKILLS AND RESOURCES | IMMEDIATE LIFE-SAVING AND BASIC NEEDS | RESTORE AND IMPROVE WHILE REDUCING FUTURE RISK

PREVENTION → avoid creation or existence of risk where possible

MITIGATION → reduce adverse impact of remaining risk → PREPAREDNESS → ready plans warnings skills and resources → RESPONSE → immediate life-saving and basic needs

SYSTEM / LEVEL / DEFINITION

- PREVENTION → avoid creation or existence of risk where possible
- MITIGATION → reduce adverse impact of remaining risk

STRUCTURE-FUNCTION MECHANISM

- PREPAREDNESS → ready plans warnings skills and resources
- RESPONSE → immediate life-saving and basic needs

SCALE / EVIDENCE LIMIT

- RECOVERY → restore and improve while reducing future risk
- PREVENTION → avoid creation or existence of risk where possible

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RECOVERY → restore and improve while reducing future risk.

03

STAGE 03 BUILD BACK BETTER LOOP

BUILD BACK BETTER LOOP | DAMAGE AND NEEDS ASSESSMENT | ESTABLISH RECOVERY BASELINE | RESTORE ESSENTIAL FUNCTIONS | REBUILD ASSETS AND SYSTEMS | RISK-INFORMED IMPROVEMENT | AVOID OLD VULNERABILITY | PREVENTION MITIGATION AND PREPAREDNESS IMPROVE

DAMAGE AND NEEDS ASSESSMENT → establish recovery baseline → REHABILITATION → restore essential functions → RECONSTRUCTION → rebuild assets and systems → RISK-INFORMED IMPROVEMENT → avoid old vulnerability → FEEDBACK → prevention mitigation and preparedness improve

SYSTEM INPUT / STARTING CONDITION

- DAMAGE AND NEEDS ASSESSMENT → establish recovery baseline
- REHABILITATION → restore essential functions

ECOLOGICAL PROCESS / TRANSFER

- RECONSTRUCTION → rebuild assets and systems
- RISK-INFORMED IMPROVEMENT → avoid old vulnerability

OUTCOME / FEEDBACK / LIMIT

- FEEDBACK → prevention mitigation and preparedness improve
- DAMAGE AND NEEDS ASSESSMENT → establish recovery baseline

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DAMAGE AND NEEDS ASSESSMENT → establish recovery baseline.

04

STAGE 04 DOMESTIC ARCHITECTURE FIREWALL

DOMESTIC ARCHITECTURE FIREWALL | DISASTER MANAGEMENT ACT | DOMESTIC LEGAL BASE | NATIONAL POLICY AND GUIDELINES | EXECUTIVE COORDINATION AND IMPLEMENTATION SUPPORT | SDMA AND DDMA | STATE AND DISTRICT STATUTORY LEVELS | ROLES COME FROM DOMESTIC LAW, NOT SENDAI TEXT

DISASTER MANAGEMENT ACT → domestic legal base

NDMA → national policy and guidelines → NEC → executive coordination and implementation support → SDMA AND DDMA → state and district statutory levels

SYSTEM / LEVEL / DEFINITION

- DISASTER MANAGEMENT ACT → domestic legal base
- NDMA → national policy and guidelines

STRUCTURE-FUNCTION MECHANISM

- NEC → executive coordination and implementation support
- SDMA AND DDMA → state and district statutory levels

SCALE / EVIDENCE LIMIT

- RULE → roles come from domestic law, not Sendai text
- DISASTER MANAGEMENT ACT → domestic legal base

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → roles come from domestic law, not Sendai text.

05

STAGE 05 TRAINING-RESPONSE SPLIT

TRAINING-RESPONSE SPLIT | TRAINING RESEARCH DOCUMENTATION CAPACITY DEVELOPMENT | SPECIALISED DISASTER RESPONSE OPERATIONS | POLICY AND GUIDANCE | DISTRICT PLANNING AND IMPLEMENTATION CONTEXT | INSTITUTION TYPE DETERMINES THE ANSWER

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
NDMI → training research documentation capacity development	NDMA → policy and guidance	RULE → institution type determines the answer
NDRF → specialised disaster response operations	DDMA → district planning and implementation context	NDMI → training research documentation capacity development

ECOLOGICAL UNIT / PARAMETER

- NDMI → training research documentation capacity development
- NDRF → specialised disaster response operations

DECISIVE DISTINCTION

- NDMA → policy and guidance
- DDMA → district planning and implementation context

SCALE / STATUS / EXCEPTION

- RULE → institution type determines the answer
- NDMI → training research documentation capacity development

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DDMA → district planning and implementation context; RULE → institution type determines the answer.

06

STAGE 06 SENDAI IDENTITY CARD

SENDAI IDENTITY CARD | 2015 TO 2030 | GLOBAL VOLUNTARY NON-BINDING FRAMEWORK | PREVENT NEW AND REDUCE EXISTING DISASTER RISK | PEOPLE LIVELIHOODS HEALTH ASSETS AND SYSTEMS | INDIAN STATUTE FUND OR RESPONSE FORCE

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
PERIOD → 2015 to 2030	PURPOSE → prevent new and reduce existing disaster risk	NOT → Indian statute fund or response force
NATURE → global voluntary non-binding framework	SCOPE → people livelihoods health assets and systems	PERIOD → 2015 to 2030

ECOLOGICAL UNIT / PARAMETER

- PERIOD → 2015 to 2030
- NATURE → global voluntary non-binding framework

DECISIVE DISTINCTION

- PURPOSE → prevent new and reduce existing disaster risk
- SCOPE → people livelihoods health assets and systems

SCALE / STATUS / EXCEPTION

- NOT → Indian statute fund or response force
- PERIOD → 2015 to 2030

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NOT → Indian statute fund or response force.

07

STAGE 07 FOUR PRIORITIES RAIL

FOUR PRIORITIES RAIL | PRIORITY 1 | UNDERSTAND DISASTER RISK | PRIORITY 2 | STRENGTHEN RISK GOVERNANCE | PRIORITY 3 | INVEST IN RESILIENCE | PRIORITY 4

PRIORITY 1 → understand disaster risk → PRIORITY 2 → strengthen risk governance → PRIORITY 3 → invest in resilience → PRIORITY 4 → preparedness response and Build Back Better → RULE → fields of action, not the seven global targets

SYSTEM INPUT / STARTING CONDITION

- PRIORITY 1 → understand disaster risk
- PRIORITY 2 → strengthen risk governance

ECOLOGICAL PROCESS / TRANSFER

- PRIORITY 3 → invest in resilience
- PRIORITY 4 → preparedness response and Build Back Better

OUTCOME / FEEDBACK / LIMIT

- RULE → fields of action, not the seven global targets
- PRIORITY 1 → understand disaster risk

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → fields of action, not the seven global targets.

08

STAGE 08 SEVEN-TARGET LOGIC

SEVEN-TARGET LOGIC | TARGETS A TO D | SUBSTANTIALLY REDUCE SPECIFIED LOSSES | MORTALITY AND AFFECTED PEOPLE | HUMAN IMPACT DIRECTIONS | ECONOMIC AND CRITICAL-SERVICE LOSS | SYSTEM IMPACT DIRECTIONS | TARGETS E TO G

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
TARGETS A TO D → substantially reduce specified losses	ECONOMIC AND CRITICAL-SERVICE LOSS → system impact directions	RULE → indicator values require official monitoring metadata
MORTALITY AND AFFECTED PEOPLE → human impact directions	TARGETS E TO G → substantially increase strategies cooperation warnings	TARGETS A TO D → substantially reduce specified losses

ECOLOGICAL UNIT / PARAMETER

- TARGETS A TO D → substantially reduce specified losses
- MORTALITY AND AFFECTED PEOPLE → human impact directions

DECISIVE DISTINCTION

- ECONOMIC AND CRITICAL-SERVICE LOSS → system impact directions
- TARGETS E TO G → substantially increase strategies cooperation warnings

SCALE / STATUS / EXCEPTION

- RULE → indicator values require official monitoring metadata
- TARGETS A TO D → substantially reduce specified losses

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → indicator values require official monitoring metadata.

09

STAGE 09 GLOBAL-DOMESTIC BRIDGE

GLOBAL-DOMESTIC BRIDGE | GLOBAL PRINCIPLES PRIORITIES AND TARGETS | INDIAN LAW | AUTHORITIES POWERS DUTIES AND FUNDS | NATIONAL AND STATE PLANS | DOMESTIC IMPLEMENTATION CHOICES | SECTOR AGENCIES | WARNINGS INFRASTRUCTURE HEALTH AND RELIEF

SENDAI → global principles priorities and targets → INDIAN LAW → authorities powers duties and funds → NATIONAL AND STATE PLANS → domestic implementation choices → SECTOR AGENCIES → warnings infrastructure health and relief → ALIGNMENT → policy bridge, not legal identity

SYSTEM / LEVEL / DEFINITION

- SENDAI → global principles priorities and targets
- INDIAN LAW → authorities powers duties and funds

STRUCTURE-FUNCTION MECHANISM

- NATIONAL AND STATE PLANS → domestic implementation choices
- SECTOR AGENCIES → warnings infrastructure health and relief

SCALE / EVIDENCE LIMIT

- ALIGNMENT → policy bridge, not legal identity
- SENDAI → global principles priorities and targets

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SECTOR AGENCIES → warnings infrastructure health and relief.

10

STAGE 10 EVIDENCE STATUS SCREEN

EVIDENCE STATUS SCREEN | LOSS FIGURE | EVENT DATE DEFINITION AND REPORTING SOURCE | TARGET PROGRESS | INDICATOR BASELINE AND PERIOD | EARLY WARNING | HAZARD GEOGRAPHY REACH AND ACCESS | INSTITUTIONAL POWER

LOSS FIGURE → event date definition and reporting source

TARGET PROGRESS → indicator baseline and period → EARLY WARNING → hazard geography reach and access → INSTITUTIONAL POWER → current statutory provision

SYSTEM / LEVEL / DEFINITION

- LOSS FIGURE → event date definition and reporting source
- TARGET PROGRESS → indicator baseline and period

STRUCTURE-FUNCTION MECHANISM

- EARLY WARNING → hazard geography reach and access
- INSTITUTIONAL POWER → current statutory provision

SCALE / EVIDENCE LIMIT

- PLAN STATUS → dated official edition or amendment
- LOSS FIGURE → event date definition and reporting source

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PLAN STATUS → dated official edition or amendment.

11

STAGE 11 DISASTER ANSWER SPINE

DISASTER ANSWER SPINE | HAZARD EXPOSURE VULNERABILITY CAPACITY RISK AND DISASTER | PREVENTION MITIGATION PREPAREDNESS RESPONSE RECOVERY | NDMA NEC NIDM NDRF SDMA AND DDMA BY OWNED | FOUR SENDAI PRIORITIES FROM SEVEN GLOBAL TARGETS | DOMESTIC LAW PLAN LOSS INDICATOR AND WARNING EVIDENCE

DEFINE → hazard exposure vulnerability capacity risk and disaster → SEQUENCE → prevention mitigation preparedness response recovery → MAP → NDMA NEC NIDM NDRF SDMA and DDMA by owned role → SEPARATE → four Sendai priorities from seven global targets → VERIFY → domestic law plan loss indicator and warning evidence

SYSTEM / LEVEL / DEFINITION

- DEFINE → hazard exposure vulnerability capacity risk and disaster
- SEQUENCE → prevention mitigation preparedness response recovery

STRUCTURE-FUNCTION MECHANISM

- MAP → NDMA NEC NIDM NDRF SDMA and DDMA by owned role
- SEPARATE → four Sendai priorities from seven global targets

SCALE / EVIDENCE LIMIT

- VERIFY → domestic law plan loss indicator and warning evidence
- DEFINE → hazard exposure vulnerability capacity risk and disaster

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → domestic law plan loss indicator and warning evidence.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | THE SENDAI FRAMEWORK AND INDIA'S EVOLVING DISASTER POLICY INCREASINGLY | ANALYTICAL SIGNIFICANCE | THIS FRAMING DIRECTLY INTEGRATES DISASTER-MANAGEMENT | INDIA'S CYCLONE EARLY-WARNING AND EVACUATION CAPABILITY (LED BY IMD | CONTINUING GAPS | EARLY-WARNING AND PREPAREDNESS CAPABILITY IS MARKEDLY STRONGER FOR | COMPARATIVE CASUALTY-REDUCTION CLAIMS ACROSS DIFFERENT DISASTER EVENTS (E.G., CITING

OPTIONAL ECOLOGICAL PROCESS DEPTH

- CAUTION: The Sendai Framework and India's evolving disaster policy increasingly recognise
- CAUTION: Analytical significance: this framing directly integrates disaster-management
- India's cyclone early-warning and evacuation capability (led by IMD forecasting combined
- CAUTION: Continuing gaps: early-warning and preparedness capability is markedly stronger for

SCALE, PARAMETER OR STATUS LIMIT

- CAUTION: Comparative casualty-reduction claims across different disaster events (e.g., citing
- CAUTION: District-level disaster-management capacity assessments (staffing, resourcing,
- CAUTION: Quantifying the specific risk-reduction value of ecosystem-based measures (e.g., exact
- The Sendai Framework represents a paradigm shift from "disaster management" (managing

COMPARATIVE RESTORATION NUANCE

- District Disaster Management Authorities (DDMAs), chaired by the District Magistrate/
- Ecosystem-based disaster-risk reduction (mangroves, wetlands, forests as risk-reduction
- India's cyclone early-warning/evacuation capability has matured substantially further
- NOT: The Sendai Framework primarily emphasises improving emergency response capability as its

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels